Artificial Economies for Learning Agents: Testing Whether Institutions Stay Robust Across Learning Architectures

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Abstract

This paper introduces a code-first experimental platform for studying whether economic mechanisms remain robust when the strategic agents inside them are adaptive learners rather than equilibrium solvers. The platform implements a shared `World/Agent/Institution` interface, repeated multiseed evaluation, exploitability tests, and interchangeable mind classes ranging from random and tabular Q-learning agents to PyTorch DQN, PPO, decorrelated independent-DQN, and centralized-critic agents. The first validated environment is a repeated two-firm pricing arena with six regulatory mechanisms. In this world, price caps robustly reduce exploitability across mind classes, but their effect on collusion is metric- and architecture-dependent. A new paired-seed audit shows that DQN's positive profit-under-cap result is not visible in 1000-step traces; it emerges with training budget, becomes mostly positive by 10,000 steps, and is positive in the 40,000-step full table. The second environment, Resource Island, extends the same interface to a spatial inventory economy and reveals a different lesson: institution conclusions are only meaningful when the world actually exercises the institutional channel. Early Resource Island versions mechanically under-tested property rights and trade price controls; a hardened v1 now creates contested resources, unequal trades, and specialization pressure. Three additional environments extend the suite: Auction House provides a benchmarked auction-design testbed, Public Goods exposes common-pool extraction and contribution incentives, and Labor Market tests learned reporting under deferred acceptance. The current contribution is both empirical and methodological: it reports initial evidence that institutional robustness changes across learning architectures, and it documents the validation discipline needed to distinguish real economic findings from inactive-mechanism or short-horizon artifacts.

1 Introduction

Economic mechanisms are usually evaluated against a model of strategic behavior: equilibrium prices, truthful bidding, efficient allocation, or some other theoretical benchmark. That evaluation is only as good as the behavioral model behind it. This paper sets out to find out what happens when institutions designed around classical economic assumptions are placed in front of learning agents; which guarantees survive, which break, and does the break depend on the learning architecture?

We answer this with five small, controlled economic worlds, a pricing duopoly (Pricing Arena), a scarce-resource trade economy (Resource Island), a sealed-bid auction (Auction House), a common-pool resource game (Public Goods), and a two-sided labor match (Labor Market), each paired with a classical benchmark and a shared ladder of learning architectures: random behavior, tabular Q-learning, deep Q-networks, proximal policy optimization, independent multi-agent Q-learning, and centralized-critic training. These architectures are not a scale from weaker to stronger. The question at stake in each world is not which architecture performs best, but which of the institution's guarantees hold up as the behavioral assumptions behind the benchmark are changed.

Across the five worlds, the same failure modes recur in different forms: an institution can look effective on one metric while a different strategic channel stays open; an apparent effect can depend on training horizon and reverse sign with more training budget; a learning architecture can fail to recover a benchmark that classical theory says is optimal; and an institution can raise a headline welfare number while quietly eroding the property, stability, sustainability, truthfulness, that the institution actually exists to protect. The next section sets out, world by world, the classical benchmark each environment owes and what claim boundary follows from it, while sections 4 through 8 report where each of these failure modes appears, and where the evidence is instead null or still provisional.

2 Related Work

This paper connects five economic literatures to a shared reinforcement-learning testbed. Each world below is built to test one specific classical prediction. The Pricing Arena is motivated by work on algorithmic collusion: repeated-game Q-learning agents can learn to sustain prices above the competitive level, without any explicit communication between them [Calvano et al., 2020, 2021]. The exploitability check used in this paper follows the same concern one step further. A policy's average performance across a run does not show whether that policy is robust to a deviating opponent, so this paper measures both. Auction House is based on classical auction theory. In a second-price auction, bidding one's true value is a dominant strategy, meaning it is optimal regardless of what other bidders do [Vickrey, 1961]. In a first-price auction, by contrast, the optimal strategy is to bid below one's true value (shading), and reserve prices create a known tradeoff between the seller's expected revenue and the efficiency of the final allocation [Myerson, 1981, Milgrom, 2004].

Public Goods and Resource Island are both based on common-pool resource and public-goods theory. The standard prediction in this literature is that, absent some form of monitoring, reputation, or locally appropriate governance, a shared resource will be overused or a public good will be underfunded, because each agent's private incentive does not account for the cost imposed on everyone else [Samuelson, 1954, Ostrom, 1990]. Labor Market is based on two-sided matching theory. Under the worker-proposing deferred acceptance algorithm, the resulting match is guaranteed to be stable (no worker and employer who are not matched to each other would both prefer to be), and truthful reporting of preferences is guaranteed to be optimal for the proposing side [Gale and Shapley, 1962, Roth and Sotomayor, 1990, Roth, 1984]. These are proven properties of the mechanism, not empirical tendencies, which is what makes them useful as benchmarks.

The learner suite is built from five standard reinforcement learning methods: tabular Q-learning, deep Q-networks (DQN) [Sutton and Barto, 2018, Mnih et al., 2015], proximal policy optimization (PPO) [Schulman et al., 2017], independent Q-learning [Tan, 1993], and centralized-critic training [Lowe et al., 2017]. What is different here is that the suite is used to test assumptions, not to compare performance. For example, second-price truthfulness and deferred-acceptance strategy-proofness are proofs about a single, fixed learner responding to a fixed mechanism; independent Q-learning and centralized-critic training instead put multiple agents in motion at the same time, which is not the setting those proofs were written for.

3 Theory Obligations

Every world in this paper owes something to the theory it is built on: a specific prediction it must be checked against, a specific set of measurements that follow from that prediction, and a specific boundary on what the resulting evidence can support. This section states that obligation for each world before any learned-agent result is discussed. The obligation exists because passing tests and producing a complete set of output files is not the same thing as having tested the institution a world claims to test.

The five worlds do not owe the same kind of proof. Grouping them by the strength of their classical prediction gives three tiers: proven mechanism properties, where a learned agent's behavior can be checked directly against a known correct answer; bracketed or reference-point theory, where exact values exist but no single predicted outcome for the object under test; and activation, where no external benchmark exists at all and the obligation is instead to show the institution was ever placed under real economic pressure.

Auction House and Labor Market belong to the first tier. In Auction House, the classical predictions are exact: truthful bidding in a second-price auction, bid shading in a first-price auction, and a known revenue-efficiency tradeoff under a reserve price. Because the correct behavior is known, this world measures distance from it directly, as revenue, allocative efficiency, bidder surplus, ex-post regret, over- and underbidding, shading distance, and the shape of the learned bid curve against the truthful or shaded reference, rather than raw payoff alone. The resulting claim is bounded to benchmark deviation under different learners; convergence to auction equilibrium is not asserted. In Labor Market, a lower truthful-report rate under some learner is not by itself evidence of profitable manipulation, given the proposing side's strategy-proofness guarantee. Match rate, blocking-pair stability, truthful-report rate, and welfare are therefore reported together, since a learner can trade one against another, and any claim about manipulation is bounded to the worker-proposing, canonical version of the mechanism implemented here.

Pricing Arena and Public Goods belong to the second tier. Pricing Arena has two exact static prices, the Nash price and the joint-profit price, but no closed-form prediction for behavior in the repeated game itself, only a range bounded by the two. A collusion claim in this world is therefore only as credible as the decomposition behind it: price, profit, quantity, welfare, and exploitability are reported together and checked against each other, and the resulting claim is limited to finite-run, regulatory evidence in a stylized duopoly, not a theorem or a real-market antitrust result. Public Goods is bracketed rather than pointed: theory predicts underprovision when private incentives diverge from the social good, with free-riding as a lower bound and the social optimum as an upper one, and nothing more specific in between. The obligation this creates is to keep the welfare number honest against the underlying resource state, since an institution can raise measured welfare through reward accounting without changing the incentive to extract or contribute at all. Contribution, extraction, sustainability, inequality, and tax revenue are therefore reported as separate quantities from welfare, so that an accounting effect is not mistaken for a real one.

Resource Island belongs to the third tier alone. It carries no external closed-form benchmark, so its obligation is not how close a learned agent gets to a known correct value, but whether the institution being measured was ever placed under real economic pressure in the first place. A property-rights rule is not being tested if no non-owner ever has reason to access a claimed resource; a trade price control is not being tested if every trade happens to be one-for-one already. Survival, sustainability, trade counts, property pressure, specialization, and institution-block counters exist to answer that prior question before any welfare number from this world is trusted. This is not a weaker test than the other four worlds run; it is a test of a condition the other four simply assume already holds, and Section 5 reports a case where that assumption failed and had to be corrected before the world's results meant anything.

4 Platform

4.1 Interfaces

The codebase is organized around three abstractions:

- A `World` defines state transitions, observations, metrics, and rendering state.
- An `Agent` or mind maps observations to actions and updates from rewards and next observations.
- An `Institution` modifies rules, rewards, constraints, or information inside a world.

This separation is not cosmetic. It makes it possible to ask whether a result is about a world, an institution, or a learner. It also creates a regression test: if a new world requires rewriting the shared learning interface, then the platform abstraction has failed.

4.2 Validation Discipline

The platform treats a result as reportable only after four checks. First, the mechanics must be unit-tested: allocation, movement, payment, trade, collision, or reward rules must match hand-computed cases. Second, where theory provides a known answer, the implementation must recover or bracket that answer before learned-policy output is interpreted. The current known-answer runner reports 19/19 passing checks across Pricing Arena N-firm static Nash references, Auction House truthfulness and bid-shading cases, Public Goods free-rider and cooperative brackets, Labor Market deferred-acceptance benchmark cases, and Resource Island gather bounds. Third, the experiment must be replicated over seeds, with per-seed and aggregate CSV outputs. Fourth, the institution must actually activate. An institution that never faces the behavior it is meant to govern produces no evidence about that institution, only about the world design. Section 5.2 gives a concrete case. The second new validation layer is mechanism tracing. The trace runner produces per-step rows and algebraic decomposition rows for all five worlds and all thesis-facing minds. These traces do not replace full-run evidence; they show the channel behind a claim and can expose contradictions between short traces and full-run results. Section 4.3 gives a concrete case.

5 Pricing Arena

5.1 World

Pricing Arena is a repeated two-firm pricing market. Each firm chooses a price from a discrete grid every round. Consumers divide between the two firms according to a noisy logit demand model, so a firm's realized quantity depends on both its own price and its rival's. Profit is price minus cost, times realized quantity, and learning agents update from repeated play against each other. Six mechanisms are implemented on top of this base game: no regulation, a price cap, a tax on high prices, random audits, a direct anti-collusion penalty, and demand shocks.

5.2 Metrics

The world reports average price, total profit, consumer surplus, welfare, price dispersion, exploitability, and two collusion indexes. The first is a price-normalized proxy used since early versions of the project. The second is a profit-normalized index, used because it maps directly onto the collusion measure common in the algorithmic-pricing literature:

$$\mathrm{CI}_{profit} = \frac{\pi_{observed} - \pi_{Nash}}{\pi_{monopoly} - \pi_{Nash}}. \tag{1}$$

Both are reported together, since the two can move in different directions, and that divergence is itself informative about which channel a mechanism is actually affecting.

5.3 Known-Answer Validation

Before any learned result is interpreted, the static Nash best response is checked directly against the world's own demand and cost parameters, for $n = 2$, 3, 4, and 5 firms. All four checks pass. This does not validate anything about learned behavior; it validates that the reference point learned behavior is measured against is not arbitrary.

5.4 Baseline and Main Result

Table 1 shows the unregulated baseline for each learner. Q-learning and the centralized critic sit at the higher-price, higher-collusion end of the range; DQN, PPO, and independent-DQN sit lower, with lower exploitability.

Mind	Avg. Price	Profit	Welfare	Collusion Index	Exploitability
Q-learning	5.487	365.3	810.4	0.541	95.7
DQN	3.611	236.3	832.8	0.202	20.3
PPO	3.681	260.0	839.3	0.215	14.2
Independent-DQN	4.078	269.9	827.9	0.285	26.6
Centralized critic	5.925	288.6	791.6	0.618	147.5

Table 1: Pricing Arena, no-regulation baseline, by learner.

Table 2 shows the change under a price cap, relative to that baseline, for every learner. The exploitability column is the clearest result in this world: it falls for every learner, without exception. The profit column is not: it falls for Q-learning and random play, in line with the conventional expectation that a cap disciplines pricing behavior, but it rises for DQN, independent-DQN, and the centralized critic.

The pattern is a guardrail that holds on the axis it was built for and does not hold on an adjacent one. A price cap constrains the price level directly, and every learner's exploitability falls as a result, since a capped price cannot be pushed into the range a deviating opponent could exploit most severely. But the cap does not constrain profit directly, only through price, and for three of the five learners the resulting change in realized quantity and market structure more than offsets the lower price. Q-learning follows the standard price-discipline expectation more closely than DQN, independent-DQN, or the centralized critic.

Mind	Price Δ	Profit Δ	Welfare Δ	Exploitability Δ
Q-learning	−0.759	−22.916	+19.567	−55.366
Random	−1.185	−16.028	+34.779	−31.775
DQN	+0.388	+55.358	+6.294	−11.619
PPO	−0.042	−6.558	−1.198	−5.704
Independent-DQN	−0.288	+3.093	+12.183	−14.871
Centralized critic	−1.337	+10.347	+28.668	−98.274

Table 2: Price cap versus no regulation, change by learner, n=20 full run.

5.5 The Training-Budget Correction

The DQN result above was not visible in the first version of this analysis. A short, 1000-step mechanism trace, run to check the channel behind the headline number, showed DQN profit falling under the cap for every seed tested, the opposite of Table 2. This discrepancy was resolved by running the same paired-seed comparison at a range of training budgets, from the short trace up to the full 40,000-step run the headline table is drawn from.

Source	Steps	Mean Profit Δ	Positive Seed Share	Mean Quantity Δ
fresh audit	1,000	−11.711	0.10	3.220
fresh audit	5,000	+1.730	0.50	4.102
fresh audit	10,000	+35.448	0.80	3.540
full table	40,000	+55.358	0.85	0.226

Table 3: DQN price-cap profit delta by training budget. Values are paired seed deltas, price cap minus no regulation.

The sign flips between 1,000 and 5,000 steps and is positive with a clear majority of seeds by 10,000. At 40,000 steps, the profit effect is statistically positive ($\Delta\pi = 55.358$, 95% CI [4.859, 105.857], positive in 85% of seeds), but the mean quantity change at that budget is small, unlike the 5,000 and 10,000-step rows where quantity moves substantially. This rules out the simplest explanation, that DQN is exploiting a fixed quantity-margin loophole from the start; the effect is not present early and does not have a single stable channel behind it at every budget. The more defensible interpretation is that the price cap becomes a focal constraint that some DQN runs learn to use only after sufficient training, rather than a channel visible from the beginning. A short trace is therefore insufficient evidence for this result; the training-budget ladder is the validation that makes the interpretation credible.

5.6 Secondary Pricing Interventions

Price cap is the only mechanism in this world that produces a consistent cross-learner pattern. The remaining four are correctly implemented and pass the same unit and parity checks as price cap, but their effects are smaller, more parameter-sensitive, or less consistent across learners. At the currently tested parameters, no single mechanism-level claim is as well supported for these interventions as it is for the price cap.

The high-price tax subtracts a quantity-weighted penalty from reward above a price threshold. Its effect is small in every direction and every learner: Q-learning's exploitability falls modestly, DQN is essentially unaffected, and the centralized critic moves in the wrong direction on price and welfare. At the tested tax rate, the penalty is too small, or too easy to avoid, to reliably change pricing behavior.

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	−0.078	+2.2	−7.6
DQN	+0.006	+0.6	+1.2
PPO	−0.023	−3.6	−5.2
Independent-DQN	+0.001	+0.2	+2.5
Centralized critic	+0.088	+4.2	−1.2

Table 4: High-price tax versus no regulation, change by learner.

Random audits apply a stochastic penalty when average price is high. The result here is not only small but directionally inconsistent: for Q-learning, exploitability rises under audits rather than falling, and for the centralized critic, profit rises sharply while exploitability falls only partially. An enforcement mechanism with opposite signs across learners does not provide a robust guardrail at these parameters.

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	+0.114	+11.1	+32.0
DQN	+0.035	+4.3	−0.5
PPO	−0.006	−1.0	−2.7
Independent-DQN	+0.002	+0.2	+1.7
Centralized critic	+0.125	+49.1	−21.7

Table 5: Random audit versus no regulation, change by learner.

The direct anti-collusion penalty punishes prices that are both high and close together, the most explicit mechanism in the suite and, on paper, the one closest to actual antitrust enforcement logic. Its observed effect is limited: four of the five learners show negligible or worsened exploitability, and only PPO shows a mild improvement across price, profit, and exploitability together. A rule aimed directly at the behavior of interest is therefore not automatically more effective than the blunter price cap in this setting.

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	−0.048	−2.1	+31.9
DQN	+0.005	+0.5	0.0
PPO	−0.067	−7.4	−5.5
Independent-DQN	+0.001	+0.2	0.0
Centralized critic	−0.013	+0.1	+0.5

Table 6: Anti-collusion penalty versus no regulation, change by learner.

Demand shocks perturb market size with a lognormal multiplier each round, testing robustness to environmental volatility rather than acting as a policy lever. Results are small and mixed in direction across learners, with the centralized critic the only one meaningfully affected, and negatively. This mechanism is best read as a stress test the platform passes mechanically, not as a candidate institutional finding.

Mind	Price Δ	Profit Δ	Welfare Δ	Exploitability Δ
Q-learning	+0.014	+4.0	+0.3	+11.4
DQN	−0.002	+1.3	+1.8	+1.3
PPO	−0.010	−0.3	+1.4	−6.0
Independent-DQN	+0.001	+0.6	+1.2	+1.8
Centralized critic	−0.050	−9.2	−12.0	+2.7

Table 7: Demand shock versus no regulation, change by learner.

These outcomes should be interpreted as economic-activation results rather than implementation defects. All four mechanisms pass the same mechanics and parity tests as price cap. At the parameters currently tested, only price cap creates pressure strong enough to produce a consistent, cross-learner, interpretable signal. Table 8 summarizes the distinction.

Mechanism	Implementation	Interpretability
Price cap	Passes all checks	Strong, consistent, cross-learner
High-price tax	Passes all checks	Weak at tested rate
Random audit	Passes all checks	Mixed, sign-inconsistent
Anti-collusion penalty	Passes all checks	Limited binding, parameter-sensitive
Demand shock	Passes all checks	Robustness check, not a policy result

Table 8: Mechanism status, implementation versus interpretability, Pricing Arena.

Pricing Arena's most interpretable result is a guardrail that succeeds on its direct target while failing to provide an invariant guarantee over adjacent economic outcomes: price caps lower exploitability for every learner tested, but their effect on profit and collusion is learner-dependent, with the DQN effect emerging only after sufficient training. The remaining four mechanisms are correctly implemented, but at current parameters their effects are too small or inconsistent to support a general claim beyond the reported values. Whether the price-cap result and the secondary-mechanism patterns persist as the number of firms increases beyond two is addressed in the cross-world synthesis section.

6 Resource Island

6.1 World

Resource Island is a bounded-grid gather-and-trade economy. Agents move across the grid, gather food and wood, maintain an energy stock that is drawn down each round, hold separate food and wood inventories, and can starve if energy runs out. Available actions are staying in place, moving in one of four directions, gathering from the current cell, and offering food for wood or wood for food in trade. The world implements property rights, redistribution, trade price controls, and a reputation system as swappable institutions on top of this base economy. It was built specifically to test whether the platform's shared `World/Agent/Institution` interface holds up outside a two-firm pricing game, in a setting with spatial structure, inventory, and multilateral exchange rather than a single simultaneous-move market.

The world records survival rate, welfare, specialization, inequality over time, resource sustainability, trade attempts, successful trades, inventory-blocked trades, institution-blocked trades, property claims, property violations, and property opportunities. This last group exists because Resource Island, unlike Pricing Arena or Auction House, has no external closed-form benchmark to check learned behavior against; its central obligation is instead to demonstrate that each institution was placed under economic pressure before any welfare or trade number from it is interpreted.

6.2 Activation as the Governing Question

Classical theory gives qualitative predictions here rather than a single target value: property rights should reduce conflict over contested resources, trade should raise welfare when agents have heterogeneous needs or access, reputation should support repeated exchange, and price controls can protect against exploitative trades while also blocking mutually beneficial ones. None of these predictions specifies a number a learned agent should reach. What they do specify is a precondition: an institution cannot be evaluated unless the world creates the situation the institution is meant to govern. A property-rights rule is not evaluated if no non-owner ever has reason to access a claimed cell, and a trade price control is not evaluated if every trade already happens to be one-for-one. This section therefore reports results only from a version of the world confirmed to place every institution under that pressure, and treats an earlier, non-activated version as a methods note rather than as economic evidence.

6.3 Activation Check

The first full runs of this world passed every unit and parity test and produced complete, statistically replicated output, but were economically under-activated. Successful trade was sparse, and property rights rarely faced behavioral pressure: even where a claim existed, non-owner opportunities to gather from claimed cells were negligible. Trade price controls could not bind because the original trade protocol only allowed strictly one-for-one exchange, which no control could flag as unequal. This was a concrete instance of an environment that was runnable, tested, and replicated across seeds, but still did not exercise the institution it claimed to study.

The corrected version, v1, adds four changes: contested resource layouts, so that multiple agents have a reason to want the same cell; unequal trade offers, so that a price control has something to bind against; specialization pressure, through heterogeneous resource preferences and starting inventories; and the activation diagnostics themselves, counting trade attempts, blocked trades, property opportunities, property violations, and institution blocks directly. All results reported below are from this activated v1 configuration, confirmed by those diagnostics to place each institution under economic pressure and validated at both a medium-scale gate run and the full $n = 20$ replication that follows it.

6.4 Baseline

Table 9 shows the no-institution baseline under Q-learning, the reference point every institution result below is measured against. Trade attempts occur roughly twice as often as successful trades, survival is high, and resource sustainability sits below one, meaning agents draw the resource stock down through ordinary activity rather than leaving it untouched.

Metric	Value
Survival rate	0.9439
Welfare	1.1065
Successful trades	5.3251
Trade attempts	10.5507
Specialization index	0.3895
Inequality over time	0.1558
Resource sustainability	0.7638

Table 9: Resource Island v1, no-institution baseline, Q-learning.

6.5 Institution Results

Property rights create a claim on a cell after its first successful gather, and can then block or penalize a later gather attempt by a non-owner. Under v1, this rule is measurably exercised: Table 10 reports nearly 28 property-opportunity events per run under Q-learning, alongside small positive changes in survival, welfare, and trade. Violations remain low relative to opportunities, so the result should not be interpreted as evidence that property rights resolve conflict in general. The supported claim is that v1 makes the mechanism observable and mildly beneficial in this configuration.

Metric	Value
Survival Δ	+0.0077
Welfare Δ	+0.0474
Trade count Δ	+0.4861
Property claims	1.8695
Property violations	0.0468
Property opportunities	27.8249

Table 10: Property rights versus no institution, Q-learning.

Trade price controls block any trade whose exchange ratio exceeds a set maximum. Table 11 shows that the control is active: successful trade falls to zero, and welfare, survival, and sustainability fall with it. The institutional channel is therefore exercised, but the observed effect is restrictive rather than welfare-improving. A rule designed to prevent unequal exchange can eliminate exchange altogether rather than making it more equitable, and that is the pattern observed in this configuration.

Reputation gives agents a reward bonus tied to a history of successful trades. Table 12 shows welfare and trade both rising under this institution. This separates the result from a purely accounting-based reward effect: trade volume also changes, so at least part of the welfare gain is associated with additional exchange.

Redistribution was tested in the earlier, non-activated configuration and is reported here only as preliminary evidence rather than an activated-world result. In that setting it slightly reduced inequality, at the cost of a small reduction in survival, welfare, and trade. Whether this pattern holds under v1 pressure has not yet been tested.

Metric	Value
Survival Δ	−0.0249
Welfare Δ	−0.1558
Trade count Δ	−5.3251
Successful trades (level)	0.0000
Institution-blocked trades	4.7862
Resource sustainability Δ	−0.0300

Table 11: Trade price controls versus no institution, Q-learning.

Metric	Value
Survival Δ	−0.0138
Welfare Δ	+0.2412
Trade count Δ	+1.3414
Trade attempt count Δ	+0.5410

Table 12: Reputation system versus no institution, Q-learning.

6.6 Cross-Mind Results

Table 13 shows the no-institution baseline across the full learner suite. The split is substantial: Q-learning, PPO, and the centralized critic all sustain meaningful trade, while DQN and independent-DQN trade much less, by roughly an order of magnitude on trade count relative to PPO. DQN and independent-DQN also show the highest resource sustainability in the table, but this should be read as reduced economic activity rather than as efficient resource governance.

Mind	Survival	Welfare	Trades	Attempts	Specialization	Sustain.
Q-learning	0.9439	1.1065	5.3251	10.5507	0.3895	0.7638
DQN	0.9102	1.0614	0.6398	1.0709	0.1790	0.9529
PPO	0.9150	1.2537	8.8919	9.8380	0.4052	0.8789
Independent-DQN	0.9064	1.0630	1.7300	2.1310	0.1524	0.9639
Centralized critic	0.9150	1.2484	8.6973	10.8525	0.4073	0.8920

Table 13: Resource Island v1, no-institution condition, full learner suite.

This is a different architecture effect from the one observed in Pricing Arena. In Pricing Arena, DQN eventually learns a profit-preserving response to a price constraint. In Resource Island, DQN and independent-DQN underuse a productive channel that is demonstrably available, since three other learners in the same environment use it extensively. The failure mode here is non-discovery of a productive market opportunity rather than exploitation of a regulatory guardrail.

Table 14 shows that this split is not confined to the baseline; it carries through to institutional activation. Property opportunities under property rights, and trade volume under reputation, both show the same DQN-and-independent-DQN-low pattern as the baseline trade numbers.

Reputation therefore amplifies an exchange channel that already has to exist for the reputation bonus to have behavioral content. For DQN and independent-DQN, the baseline trade channel is limited, so the reputation effect is also limited. Trade price controls show the reverse pattern: successful trade falls to zero for every learner under this institution, so the formal rule binds universally, but the amount of blocked-trade pressure differs by learner, from 4.79 for Q-learning

Mind	Property Opportunities	Trade (Reputation)	Welfare (Reputation)
Q-learning	27.8249	6.6665	1.3477
DQN	8.2279	2.4212	1.1686
PPO	27.1268	13.9870	1.8244
Independent-DQN	6.5695	2.2882	1.1541
Centralized critic	24.9577	7.7649	1.4858

Table 14: Institution activation by learner, property rights and reputation.

to 0.04 for PPO. The practical economic effect of the same rule therefore depends on how much a given learner attempts to trade in the first place.

6.7 What Remains Open

The DQN and independent-DQN trade result above is reported at a single training budget, and Pricing Arena's price-cap result showed why single-budget interpretations require caution. It is possible that DQN discovers the trade channel more slowly than PPO or the centralized critic rather than failing to discover it altogether. Until a budget ladder analogous to Table 3 is run for this world, the claim is limited to the training budget actually tested.

Resource Island's contribution is different in kind from Pricing Arena's. Its early configuration demonstrated a methodological failure mode: an environment can pass mechanics tests and still fail to exercise the institution it claims to study. The activated v1 configuration then produced an institutional result: trade price controls bind and eliminate exchange, reputation amplifies trade where a learner already engages in it, and property rights become measurable rather than purely nominal. The learner-architecture split is consistent across the baseline and the institutions tested: Q-learning, PPO, and the centralized critic sustain meaningful trade economies, while DQN and independent-DQN do not at the training budget tested here.

7 Auction House

7.1 World

Auction House is a single-item sealed-bid auction world with independent private values, discrete bids, deterministic tie-breaking, first-price and second-price payment rules, and optional reserve prices [Vickrey, 1961, Myerson, 1981, Milgrom, 2004]. Each bidder observes its own valuation bin and chooses a bid. The world reports seller revenue, bidder surplus, total welfare, allocative efficiency, regret, overbidding, underbidding, and distance from benchmark bid functions.

7.2 Benchmarks

The second-price benchmark is truthful bidding [Vickrey, 1961]. The first-price benchmark is bid shading under symmetric independent private values, with discrete-grid regret checks used where the continuous benchmark is too coarse [Milgrom, 2004, Dütting et al., 2019, Banchio and Skrzypacz, 2022]. Reserve-price variants evaluate the revenue-efficiency tradeoff against the no-reserve benchmark [Vickrey, 1961, Myerson, 1981, Milgrom, 2004].

7.3 Validation

The Auction House validation run is economically coherent but not yet final evidence. Second-price learning moves toward the truthful benchmark, first-price learning shows underbidding and bid shading, and reserve prices increase revenue while changing allocative efficiency. Table 15 summarizes the current validation run.

Scenario	Revenue	Welfare	Efficiency	Underbid Rate
second price	3.373	6.216	0.747	0.381
first price	3.481	6.331	0.768	0.774
second price + reserve	4.031	6.098	0.814	0.366

Table 15: Auction House validation results.

7.4 Cross-Mind Auction Diagnostics

The full P.6 auction learner-suite run confirms that the auction world should be read against benchmark deviations, not raw reward alone. In second-price auctions, PPO has the lowest ex-post regret among the tested learning minds (0.187) and higher allocative efficiency than Q-learning (0.762 versus 0.734). DQN and fixed independent-DQN earn higher seller revenue but with lower allocative efficiency. The centralized-critic scaffold performs poorly in this asymmetric bidding setting, with low revenue and high regret.

First-price auctions show the expected underbidding/shading channel across the non-centralized learners [Milgrom, 2004]. Q-learning, DQN, PPO, and fixed independent-DQN all underbid frequently, with first-price underbid rates between 0.716 and 0.814 [Milgrom, 2004]. This makes Auction House theory-anchored: the economic question is how each learner deviates from truthful second-price bidding or shaded first-price bidding, not whether the environment merely produces positive payoffs [Vickrey, 1961, Milgrom, 2004, Dütting et al., 2019].

8 Public Goods

8.1 World

Public Goods is a repeated common-pool resource environment, following the standard tension between private extraction and shared surplus [Samuelson, 1954, Hardin, 1968]. Agents choose between contributing to the pool, extracting from it, or remaining inactive. Extraction pays private reward but depletes the shared stock; contribution costs the contributor but helps sustain the pool. When requested extraction exceeds available stock, extraction is rationed proportionally. Deterministic regeneration makes the sustainability channel easy to audit.

The implemented institutions are no intervention, a penalty schedule, contribution matching, reputation rewards, information restriction, and a generic tax-and-redistribution schedule. The main metrics are welfare, sustainability, total contribution, extraction relative to regeneration, inequality, collapse diagnostics, and tax revenue where applicable.

8.2 Q-Learning Full Run

The n=20 tabular Q-learning full run confirms the intended commons pressure. In the baseline, agents mostly extract and contribute little: sustainability is 0.0892 and total contribution is 0.1418.

Contribution matching improves both welfare and sustainability in this setting, reaching welfare 2.0808 and sustainability 0.1048. Reputation is a strong reward-shaping intervention, with welfare 9.5750, while the tax schedule mainly changes redistribution/accounting at the tested rates.

The effect validator classifies the tax schedule as mostly reward/accounting under this configuration, while penalty, contribution matching, reputation, and information restriction change state metrics relative to baseline. This is a diagnostic classification, not yet a final welfare ranking.

8.3 Cross-Mind Status

The full Public Goods learner suite shows that stronger function approximation or centralized training does not automatically solve the commons problem. Under no institution, Q-learning reaches welfare 1.815, sustainability 0.089, and contribution 0.142. DQN and fixed independent-DQN are close but contribute less; PPO and centralized-critic contribute almost nothing and sit near the lower sustainability regime.

Contribution matching is the clearest state-changing intervention in the full learner-suite run. It improves Q-learning sustainability from 0.089 to 0.105 and fixed independent-DQN sustainability from 0.085 to 0.121, with a larger increase in contribution for independent-DQN. It does not rescue PPO in this configuration. The economic reading is therefore conditional: the institution can improve the state variables when the learner explores contribution, but a more capable policy class can still converge to low-contribution behavior.

9 Labor Market

9.1 World

Labor Market is a two-sided matching environment. Workers are learning agents; employers have fixed preferences. Each worker reports a top choice, the world constructs reported preferences, and matching is resolved using deferred acceptance [Gale and Shapley, 1962, Roth and Sotomayor, 1990]. The world reports match rate, total welfare, truthful-report rate, stability, blocking-pair diagnostics, and manipulation-gain diagnostics.

This world is intentionally asymmetric. Unlike Pricing Arena, Resource Island, Public Goods, and Auction House, not every economic actor is a learning agent. That asymmetry is the main interface test for the learning-architecture suite: DQN, PPO, independent-DQN, and centralized-critic operate only over the worker side while employers remain fixed preference holders.

9.2 Q-Learning Full Run

The n=20 tabular Q-learning full run is substantially more stable than the short smoke run. Match rate remains `1.0`, while `stability_mean=0.9508`, `truthful_report_rate_mean=0.7859`, and `total_welfare_mean=3.6393`. The report-top action space creates truthfulness variation, but worker-proposing deferred acceptance still produces mostly stable matchings under the current random-preference setup.

Fixed benchmark cases clarify the interpretation. Under worker-proposing deferred acceptance, workers should not have profitable report deviations in the canonical strategy-proof setting [Gale and Shapley, 1962, Roth and Sotomayor, 1990, Roth, 1984]. Future manipulation tests therefore need to target the non-proposing side, change the mechanism, or add information and commitment frictions rather than treating worker misreports under standard deferred acceptance as the main expected failure mode.

9.3 Cross-Mind Status

The full Labor Market learner-suite run keeps match rate at 1.0 for every mind, so the relevant margins are stability and truthful-report behavior. DQN slightly improves stability relative to Q-learning (0.975 versus 0.951) while reporting truthfully less often. PPO and fixed independent-DQN preserve high stability but also lower truthfulness. The centralized-critic scaffold has the weakest matching behavior in this asymmetric setup, with stability 0.749 and truthful-report rate 0.433.

Economically, this is consistent with the benchmark warning: worker-proposing deferred acceptance is strategy-proof for workers in the canonical setting, so lower truthful-report rates are not automatically evidence of profitable manipulation [Gale and Shapley, 1962, Roth and Sotomayor, 1990, Roth, 1984]. The result is better framed as a learned-reporting diagnostic: the mechanism keeps matches mostly stable for most learners, while the centralized-critic architecture is sensitive to this asymmetric environment.

10 Cross-World Synthesis

The emerging thesis is not that one institution always works, or that the agent classes form a universal intelligence ranking. The stronger claim is more specific: institutional robustness must be evaluated jointly with the learning architecture, the training horizon, and activation diagnostics. Tabular value learning, function approximation, on-policy policy gradients, decorrelated independent learners, and centralized critics violate different assumptions of the classical benchmark story. The comparisons are therefore best read as a learner-architecture stress test, not as a scalar capability ranking.

The Pricing Arena result shows architecture-sensitive institutional effects: price caps reduce exploitability broadly, but DQN's profit response flips sign with training budget and is positive in the full paired audit. Resource Island shows the complementary failure mode: if the world design does not put an institution under pressure, then the absence of an effect is not a mechanism result. Auction House adds theory-anchored bidding benchmarks [Vickrey, 1961, Myerson, 1981]. Public Goods adds a sustainability and commons-collapse channel. Labor Market adds a matching-stability and truthfulness channel.

The five-world synthesis is now built from full n=20 learner-suite outputs for Pricing Arena, Resource Island v1, Auction House, Public Goods, and Labor Market. The protocol-comparability report passes for all five worlds, and the publication inference table includes all worlds under the same confirmatory bookkeeping. The claim-audit suite adds a second pass over those outputs: paired seed effects, benchmark gaps, metric-conflict warnings, activation confirmations, and trace/full-run sign agreement where paired traces exist. In its current run, the audit produces 1,025 paired or benchmark rows and 31 metric-conflict or activation rows. This audit layer prevents the cross-world synthesis from becoming a table of averages without claim boundaries.

The main synthesis claim is qualitative but concrete. Learning architecture changes institutional outcomes in every world, but not through one universal direction. In Pricing Arena, DQN's price-cap profit increase is a long-training, seed-heterogeneous effect rather than a short-run channel. In Resource Island v1, PPO and centralized-critic activate trade strongly while DQN-family learners remain more conservative [Mnih et al., 2015, Schulman et al., 2017]. In Auction House, the relevant question is distance from truthful, shaded, or reserve-price benchmarks rather than raw reward [Vickrey, 1961, Myerson, 1981, Milgrom, 2004]. In Public Goods, the relevant state variables are contribution, extraction, and sustainability, while some tax effects are accounting effects rather than state changes. In Labor Market, deferred acceptance produces high match rates and mostly stable outcomes, so future manipulation claims must target the right side of the market or a different matching institution.

11 Limitations

Several limitations are deliberate. The environments are small and stylized. They are designed for auditability rather than realism. The deep-RL agents are structured-observation MLP agents rather than large-scale policies. The Resource Island trade protocol is still simple, and the current v1 pressure configuration is an activation test rather than a realistic market. Auction House currently has benchmarked sealed-bid full runs and smoke-tested clock and information variants. Public Goods is still a stylized common-pool model rather than an empirical commons. Labor Market currently uses worker-side learning under deferred acceptance, so it tests one asymmetric matching channel rather than matching-market design generally.

The most important limitation is interpretive. A complete CSV does not imply a validated economic result. Every institution must be checked for activation. Every learner-architecture comparison must be checked for implementation artifacts. The independent-DQN alias issue is a concrete example: the table was statistically valid as an output file, but not valid as evidence for a distinct mind until the implementation was corrected and rerun.

The Pricing Arena price-cap audit adds a second interpretive warning. A mechanism trace can reveal the local channel but cannot replace a replicated training-budget audit. The 1000-step trace correctly showed early profit losses under the cap; the 40,000-step table correctly showed a positive DQN cap-minus-none profit delta. The result is reportable only because the contradiction was resolved through paired seed deltas and a budget ladder.

12 Reproducibility

The repository records validation status in `ROADMAP_STATUS.md`. Core outputs are stored under `outputs/`. Major runners include:

- `run_multiseed.py` for Pricing Arena replicated runs;
- `run_exploitability.py` for frozen-policy adversarial evaluation;
- `run_phase3_full.py` and `validate_phase3_full.py` for full Phase 3 Pricing Arena checks;
- `run_known_answer_sanity_checks.py` for theorem/bracketing checks before learned-policy interpretation;
- `run_mechanism_traces.py` for per-step mechanism traces and decomposition tables across worlds;
- `run_pricing_quantity_margin_audit.py` for the Pricing Arena price-cap budget-ladder contradiction audit;
- `run_claim_audit_suite.py` for cross-world paired-effect, benchmark-gap, metric-conflict, activation, and trace-alignment audits;
- `run_resource_island_smoke.py` for Resource Island smoke, validation, and full runs;
- `run_auction_house_smoke.py` for Auction House runs;

- `run_public_goods_smoke.py` for Public Goods runs;
- `run_labor_market_smoke.py` for Labor Market runs;
- `build_world_mind_comparison.py` and `build_cross_world_synthesis.py` for ladder comparisons and cross-world synthesis.

The test suite currently includes unit, integration, and output-validation coverage for the core abstractions, Pricing Arena, Resource Island, Auction House, Public Goods, Labor Market, and deep-RL/MARL minds.

12.1 Primary Output Artifacts

The main numerical claims in the paper are backed by the following generated artifacts:

- `outputs/known_answer_sanity_checks_current/summary.csv`;
- `outputs/mechanism_traces_local_full/mechanism_decomposition.csv`;
- `outputs/phase3_full/mind_comparison.csv`;
- `outputs/pricing_quantity_margin_audit/dqn_budget_ladder.csv`;
- `outputs/pricing_quantity_margin_audit/existing_full_seed_deltas.csv`;
- `outputs/resource_island_v1_full/summary_aggregate.csv`;
- `outputs/resource_island_v1_strict_radius_full/summary_aggregate.csv`;
- `outputs/resource_island_v1_phase3_full/mind_comparison.csv`;
- `outputs/auction_house_phase3_full/mind_comparison.csv`;
- `outputs/public_goods_phase3_full/mind_comparison.csv`;
- `outputs/labor_market_phase3_full/mind_comparison.csv`;
- `outputs/cross_world_synthesis/synthesis_table.csv`;
- `outputs/claim_audit_suite/claim_audit_report.md`.

13 Conclusion

This draft reports a first version of an artificial-economies platform for testing institutional robustness under learning agents. The mature Pricing Arena result already shows why the question matters: exploitability and collusion can move differently under the same institution, and a headline effect can depend on training budget rather than appearing in short traces. Resource Island, Auction House, Public Goods, and Labor Market extend the project beyond pricing games. Their role is not merely to add more checkboxes, but to test whether the patterns found in Pricing Arena survive when the economic structure changes and when the relevant institutional channel is actually activated.

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